Module_2: *Notebook*

Team Members:

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Project Title:

The Extent of Fibrosis in the Lung at different depths

Project Goal:

This project seeks to analyze developed images in order to predict the extent of fibrosis in the lung at different biopsy depths from the top of the lung.

Disease Background:

- Prevalence & incidence

  Prevalence = the number of cases of a disease in a specific population at a particular timepoint or over a set period of time. Formula:

  Prevalence = number of existing cases at one time/ total population at the same time

  For pulmonary fibrosis, prevalence in the United States ranges from about 14 to 64 cases per 100,000 people, depending on how the disease is defined. Globally, rates are within a similar range. The rates can vary by country and study. Overall, pulmonary fibrosis is rare but appears to be increasing slightly as diagnostic methods improve.

  Incidence = the rate of new cases of a disease occurring in a specific population over a certain period of time. Formula:

  Incidence = number of new cases during a time period/ population at the start of the time period For idiopathic pulmonary fibrosis, incidence is estimated at around 7 to 17 new cases per 100,000 people each year in the U.S., with similar rates observed internationally. This means that while the disease remains uncommon, new diagnoses continue to occur each year, particularly among older *adults.*

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  cite: https://s4be.cochrane.org/blog/2020/11/06/
prevalence-vs-incidence-what-is-the-difference/
  cite: https://pmc.ncbi.nlm.nih.gov/articles/
PMC9487229/?utm_source=chatgpt.com
  cite: https://respiratory-research.biomedcentral.com/
articles/10.1186/s12931-021-01791-z?utm_source=chatgpt.com
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- Risk factors (genetic, lifestyle)

  Pulomonary fibrosis is likely to affect middle-aged and older adults. Although, fibrosis in connective tissue disease is commonly seen in younger people.

  Factors that can raise your risk of pulmonary fibrosis include:

  Smoking: If a patient smokes or used to smoke, it causes a higher risk of pulmonary fibrosis than people who never smoked. Patients with emphysema are at higher risk, as well.

  Certain types of work: Patients have a higher risk of developing pulmonary fibrosis if they work in mining, farming or construction. The risk also increases if patients have continuous/repeated contact with pollutants known to damage the lungs.

  Cancer treatments: Having radiation treatments to the chest or using certain chemotherapy medicines can raise the risk of developing pulmonary fibrosis.

  Genetics. Some types of pulmonary fibrosis run in families, so genes may play a role.

  Cite: https://www.mayoclinic.org/diseases-conditions/pulmonary-fibrosis/symptoms-causes/syc-20353690

- Symptoms:

  Shortness of breath. Dry cough. Extreme tiredness. Weight loss that's not intended. Aching muscles and joints. Widening and rounding of the tips of the fingers or toes, called clubbing.

  Pulmonary fibrosis can worsen with time, and the severity of the symptoms can vary with each patient.

  cite: https://www.mayoclinic.org/diseases-conditions/pulmonary-fibrosis/

symptoms-causes/syc-20353690

- Standard of care treatment(s)

  Medicines: Healthcare professionals may recommend pirfenidone (Esbriet) or nintedanib (Ofev) in order to treat idiopathic pulmonary fibrosis. These drugs have been approvded by the Food and Drug Administration (FDA). Nintedanib is used for conditions of idiopathic pulmonary fibrosis that develop quickly. Additionally, nintedanib causes the side effects: diarrhea and nausea. On the other hand, pirfenidone can cause side effects including: nausea, loss of appetite and skin rash from sunlight. Regular blood tests are completed to check how well the liver is working.

  Oxygen therapy: Oxygen therapy is the usage of supplemental oxygen. This is done to make breathing and exercise easier. It can prevent complications due to low blood oxygen levels. It can also decrease pain from the right side of the heart. lastly, it can improve one's sleep and well-being overall. The supplementary oxygen can be carried in a small tank or a portable oxygen concentrator.

  Pulmonary rehabilitation: Pulmonary rehabilitation can assist with symptoms and improving the ability to complete daily tasks. Pulmonary rehabilitation programs focus on:

  Physical exercise to improve endurance. Breathing techniques to improve lung capacity. Nutritional counseling. Emotional counseling and support. Education about your condition.

  cite: https://www.mayoclinic.org/diseases-conditions/pulmonary-fibrosis/diagnosis-treatment/drc-20353695

- Biological mechanisms (anatomy, organ physiology, cell & molecular physiology)

  Idiopathic pulmonary fibrosis (IPF) is related to aberrant repair after repeated alveolar epithelial micro-injuries. This can lead to excessive extracellular matrix (ECM) deposition and lung scarring.

  Key mechanisms include:

    - *epithelial cell dysfunction*
    - *activation of mesenchymal cells (fibroblasts)*
    - *dysregulated immune responses (especially macrophages)*

- *pro-fibrotic signaling pathways (TGF-β)*

Aspects that can initiate the disease by influencing the core cellular and molecular events:

- Genetic predispositions
- epigenetic changes
- environmental factors (like smoking)

cite: https://pmc.ncbi.nlm.nih.gov/articles/PMC8142468/#:~:text=The%20complexity%20of%20IPF%20biology,(Riche

Data-Set:

The dataset used in this project was obtained from Dr. Shayn Peirce-Cottler's laboratory at the University of Virginia. The dataset comprises 76 images taken at different depths of the fibrotic lungs of mice. In this data, white particles indicate a fibrotic lesion. The mice were given lung fibrosis manually through the injection of the drug bleomycin. After three weeks, the mice were humanely sacrificed, and their lungs were harvested. After this, the lungs were fixed with paraformaldehyde, mounted with gel/wax, and then sliced transversely using a cryotome. An immuostain was then added to each slice after being placed on a microscopic slide. This slide was then placed under a microscope, and an image was generated. The images were then focused to include the desmin signalling for the sake of the project.

Data Analysis:

The initial part of this code taxes six randomly selected pictures from the data set and calculating the percentage of white pixels comprising the image to indicate fibrosis. This, along with the file name and depth (as provided) are then added to a CSV file to be further analyzed. The next step in the code is to generate an interpolation of the data (linear and quadratic) through Python data analysis. The accuracy of the interpolation was then analyzed and the best fit was chosen using a sample point at 3900 microns.

In [8]:
```
'''Module 2: count black and white pixels in a .jpg and extrapolate points'''

from termcolor import colored
import cv2
import numpy as np
import matplotlib.pyplot as plt
from scipy.interpolate import interp1d
```

```
import pandas as pd

# Load the images you want to analyze

filenames = [
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658
    r"C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658
            ]

# Enter the depth of each image (in the same order that the images are listed
depths = [
    45,
    1500,
    7200,
    8100,
    7000,
    330
    ]

# Make the lists that will be used

images = []
white_counts = []
black_counts = []
white_percents = []

# Build the list of all the images you are analyzing

for filename in filenames:
    img = cv2.imread(filename, 0)
    images.append(img)

# For each image (until the end of the list of images), calculate the number c

for x in range(len(filenames)):
    _, binary = cv2.threshold(images[x], 127, 255, cv2.THRESH_BINARY)

    white = np.sum(binary == 255)
    black = np.sum(binary == 0)

    white_counts.append(white)
    black_counts.append(black)

# Print the number of white and black pixels in each image.

print(colored("Counts of pixel by color in each image", "yellow"))
for x in range(len(filenames)):
    print(colored(f"White pixels in image {x}: {white_counts[x]}", "white"))
    print(colored(f"Black pixels in image {x}: {black_counts[x]}", "black"))
```

```
    print()

# Calculate the percentage of pixels in each image that are white and make a l

for x in range(len(filenames)):
    white_percent = (100 * (white_counts[x] / (black_counts[x] + white_counts[
    white_percents.append(white_percent)

# Print the filename (on one line in red font), and below that line print the

print(colored("Percent white px:", "yellow"))
for x in range(len(filenames)):
    print(colored(f'{filenames[x]}:',"red"))
    print(f'{white_percents[x]}% White | Depth: {depths[x]} microns')
    print()

'''Write your data to a .csv file'''

# Create a DataFrame that includes the filenames, depths, and percentage of wh
df = pd.DataFrame({
    'Filenames': filenames,
    'Depths': depths,
    'White percents': white_percents
})

# Write that DataFrame to a .csv file

df.to_csv('Percent_White_Pixels.csv', index=False)

print("CSV file 'Percent_White_Pixels.csv' has been created.")
```

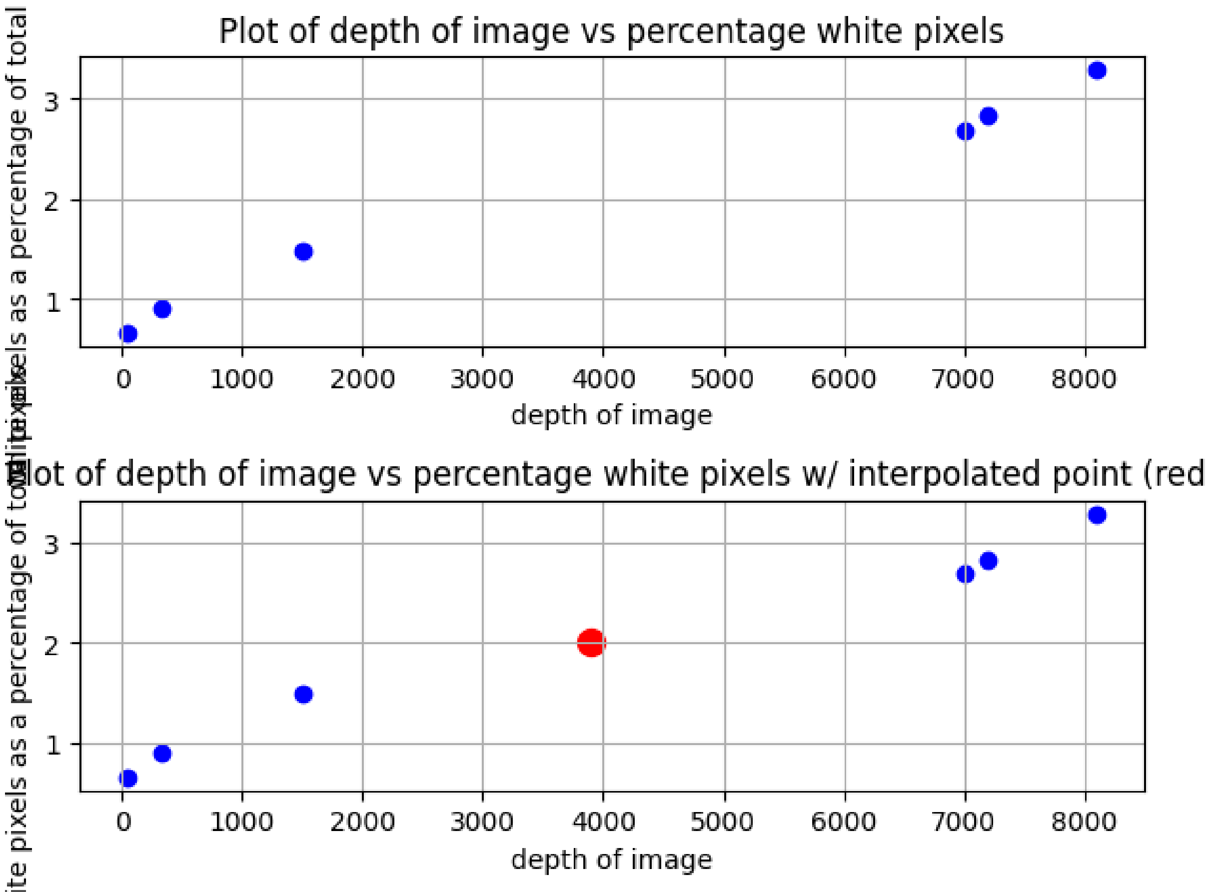
```
Counts of pixel by color in each image
White pixels in image 0: 27561
Black pixels in image 0: 4166743

White pixels in image 1: 62508
Black pixels in image 1: 4131796

White pixels in image 2: 118409
Black pixels in image 2: 4075895

White pixels in image 3: 137688
Black pixels in image 3: 4056616

White pixels in image 4: 112613
Black pixels in image 4: 4081691

White pixels in image 5: 38068
Black pixels in image 5: 4156236

Percent white px:
C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658 Llobe c
h010017.jpg:
0.6571054458618164% White | Depth: 45 microns

C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658 Llobe c
h010067.jpg:
1.4903068542480469% White | Depth: 1500 microns

C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_Sk658 Llobe c
h010160.jpg:
2.8230905532836914% White | Depth: 7200 microns

C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658 Slobe c
h010105.jpg:
3.2827377319335938% White | Depth: 8100 microns

C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658 Slobe c
h010130.jpg:
2.684903144836426% White | Depth: 7000 microns

C:\Users\rmslo\OneDrive\Documents\GitHub\project-two-compbme\MASK_SK658 Slobe c
h010156.jpg:
0.9076118469238281% White | Depth: 330 microns

CSV file 'Percent_White_Pixels.csv' has been created.
```

In [12]:

```
# Linear Interpolation:

interpolate_depth = float(input(colored("Enter the depth at which you want to 

x = depths
y = white_percents

i = interp1d(x, y, kind='linear')  # You can also use 'quadratic', 'cubic', et
```

```
interpolate_point = i(interpolate_depth)
print(colored(f'The interpolated point is at the x-coordinate {interpolate_dep

depths_i = depths[:]
depths_i.append(interpolate_depth)
white_percents_i = white_percents[:]
white_percents_i.append(interpolate_point)

# make two plots: one that doesn't contain the interpolated point, just the da
fig, axs = plt.subplots(2, 1)

axs[0].scatter(depths, white_percents, marker='o', linestyle='-', color='blue'
axs[0].set_title('Plot of depth of image vs percentage white pixels')
axs[0].set_xlabel('depth of image')
axs[0].set_ylabel('white pixels as a percentage of total pixels')
axs[0].grid(True)


axs[1].scatter(depths_i, white_percents_i, marker='o', linestyle='-', color='b
axs[1].set_title('Plot of depth of image vs percentage white pixels w/ interpo
axs[1].set_xlabel('depth of image')
axs[1].set_ylabel('white pixels as a percentage of total pixels')
axs[1].grid(True)
axs[1].scatter(depths_i[len(depths_i)-1], white_percents_i[len(white_percents_


# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()
```

```
The interpolated point is at the x-coordinate 3900.0 and y-coordinate 2.0115852
35595703.
```

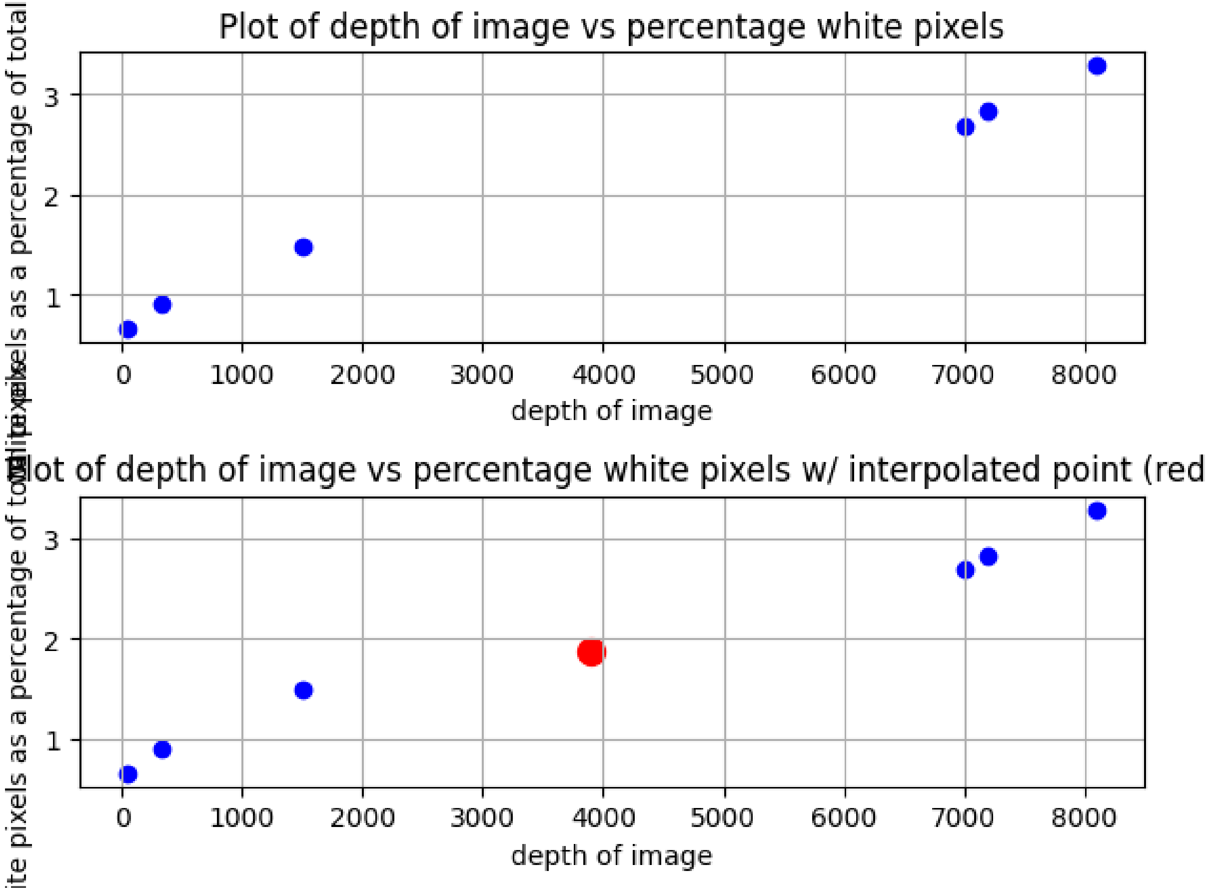


In [13]:

```
# Quadratic Interpolation:

x = depths
y = white_percents

i = interp1d(x, y, kind='quadratic')  # You can also use 'linear', 'cubic', et
interpolate_point = i(interpolate_depth)
print(colored(f'The interpolated point is at the x-coordinate {interpolate_dep

depths_i = depths[:]
depths_i.append(interpolate_depth)
white_percents_i = white_percents[:]
white_percents_i.append(interpolate_point)

# make two plots: one that doesn't contain the interpolated point, just the da
fig, axs = plt.subplots(2, 1)

axs[0].scatter(depths, white_percents, marker='o', linestyle='-', color='blue'
axs[0].set_title('Plot of depth of image vs percentage white pixels')
axs[0].set_xlabel('depth of image')
axs[0].set_ylabel('white pixels as a percentage of total pixels')
axs[0].grid(True)


axs[1].scatter(depths_i, white_percents_i, marker='o', linestyle='-', color='b
axs[1].set_title('Plot of depth of image vs percentage white pixels w/ interpo
```

```
axs[1].set_xlabel('depth of image')
axs[1].set_ylabel('white pixels as a percentage of total pixels')
axs[1].grid(True)
axs[1].scatter(depths_i[len(depths_i)-1], white_percents_i[len(white_percents_


# Adjust layout to prevent overlap
plt.tight_layout()
plt.show()
```

```
The interpolated point is at the x-coordinate 3900.0 and y-coordinate 1.8724629
960846566.
```



Based on the two types of interpolations presented, each one supports the end conculsion that increased depth leads to increased presence of white pixels (fibrosis). However, due to the quadratic graph being able to account for more unpredictability for the depth, the fit is much better.

Verify and validate your analysis:

As previously mentioned above, both of the graphs show a relationship where an increased depth of image results in an increased presence of white pixels, which indicates increased presence of idiopathic lung fibrosis. When looking at both of *the graphs, the y values make sense through this relationship. For both types of interpolations, the y values were in very similar places, lining up well with their respective existing plots.*

To validate the accuracy of this anaylsis, a test file was created to analyze the image from the data set actually taken at 3900 microns. Through the ananysis outlined in the beginning of the code, image 10065 (at 3900 microns) was 2.0015478134155273% white. The linear interpolation's graph was 2.011585235595703% while and the quadratic interpolation's graph was 1.8724629960846566%. Both of these interpolations are fairly close to the actual value (especially the linear graph).

Conclusions and Ethical Implications:

The analysis of lung fibrosis images at varying depths shows a relationship with image depth and fibrotic tissue presence. The linear and quadratic interpolation graphs demonstrate an upward trend. This supports the idea of deeper lung sections containing a higher percentage of whilte pixels (fibrotic lesions). The linear model gave an estimation of 2.01% vs. 2.00% white pixels at 3900 microns. The quadratic model was needed because it shows biological variability in the lung tissue structure.

The finding can lead to the conclusion that fibrosis severity increases with depth. This could be consistent with physiological expectations of the progressive scarring in deeper lung tissue regions. This supports the physiological understanding of how lung fibrosis develops spatially. This can make refinements wtih biopsy depth in research/diagnostics.

Analyzing the ethical implications, this data analysis supports the importance of using animal models responsibly. The data came from mouse lungs treated with bleomycin. THis is necessary in order to comply with humane research standards. Addtionally, researchers should avoid overgeneralized findings from animal models with humnan disease without tranlational studies. Quantitative image analysis can reduce animal use by improving fibrosis measurement accuracy. This uses the 3R principles of Replacement, Reducution, and Refinement in BME research.

Limitations and Future Work:

The results were consistent and biologically reasonable. From this, limitations can be made.

1. Limited Sample Size: Only 6 out of 76 available images were analyzed. This restricts statistical significances and may not fully represent the variablility across the dataset.

2. Simplified Image Analysis: The binary threshold method (white and black pixels) can oversimplify tissue complexity and introduce classification bias.

3. Interpolation Assumption: Linear and quadratic interpolations assume smooth trends with depth. However, real biological data may include abrupt regional differences.

4. Model Translation: Using a mouse fibrosis model limits direct applicability to human idiopathic pulmonary fibrosis

Future work should address the limitations by:

1. Expanding dataset to include all images for stronger statistical analysis.

2. Applying advanced segmentation methods like machine learning for color deconvolution algorithms. This can give more accurate fibrosis quantificaion

3. Performing 3D reconstructions on the images to show spatial fibrosis and its progression.

4. Comparing the fibrotic trends with different experimental conditions/treatments to convey test therapeutic impacts

5. Using statistical testing/regression models to strengthen correlation between depth and fibrosis percentage.

NOTES FROM YOUR TEAM:

This is where our team is taking notes and recording activity.

Date: 10/2/25

- *Ryan created the repository for our code and files. He downloaded the terminals for the code. Hazel got her github to be locally located on her computer. She was able to open her github folder into VSC. The team has decided to work on the project individually before next Tuesday.*

Date: 10/7/25

- *Ryan completed the code for the data analyzation. He pushed it to the github repository. Hazel completed the project title and the background* *information for check-in 1 for the project. Ryan will complete the data-set portion of the notebook after class. Hazel will complete the notes and questions for the TA. Hazel figured out how to push code to the repository!*

Date: 10/9/25

- *Ryan finished the linear and quadratic interpolation and deduced the best interpolation type was quadratic. Ryan updated the questions and notes for TAs section after today's class. Ryan also finished the verification and validation of the data and will try and find external literature to verify the findings of the analysis*

QUESTIONS FOR YOUR TA:

These are questions we have for our TA.

Date: 10/2/25

- *The team had questions about how to open github locally after Ryan created the repository. We also had a question about how the repository worked.*

Date: 10/7/25

- *The team had quesitons about how to push code to the repository. Hazel figured out how to do it after class*

Date: 10/9/25

- *No questions for the TAs*